I am applying for the Research Engineer position in the CARE-AI project because it sits at the intersection of healthcare, machine learning, and real-world implementation. As I complete my Master's in Data Science at Uppsala University, I have increasingly focused my studies and projects on healthcare applications of data and AI. I am considering pursuing a PhD in this area, and a chance to work on a project beforehand would help me understand what it looks like and the kind of PhD I should apply for. This role offers exactly that opportunity.

My interest in healthcare began long before my master's studies. Growing up in a hospital compound due to my father's work exposed me to healthcare environments from an early age. During my studies, I became interested in how machine learning could be applied in healthcare and, equally importantly, where its limitations lie. One of the first pieces I wrote on the topic explored how biases present in healthcare data and literature could be amplified through AI systems if not carefully evaluated. Since then, I have intentionally pursued projects that helped me better understand the healthcare ecosystem from both technical and operational perspectives.

Through a health innovation course, I studied the European health technology landscape and the process of bringing healthcare innovations into practice. In another project, I worked with healthcare data flows and interoperability challenges in Sweden, examining the gap between existing standards and how information is exchanged in practice. Very recently, I began a pilot collaboration with the Haematology and Oncology Department at King Faisal Specialist Hospital in Saudi Arabia, exploring AI-assisted interpretation of electrophoresis results to reduce laboratory turnaround times. These experiences have given me an appreciation for the fact that successful healthcare AI depends not only on model performance, but also on data quality, system integration, governance, and stakeholder collaboration.

My professional experience has also prepared me for the practical aspects of this position, being the bridge between technical and implementation. At Samsung Electronics, I worked extensively in ETL, with data quality, process improvement, automation, and cross-functional analysis. Much of the work involved understanding how data moved through different systems, identifying bottlenecks, improving workflows and supporting stakeholders with different technical backgrounds. While the domain was different, the underlying challenges of maintaining reliable data pipelines and ensuring that systems support decision-making are highly transferable.

Technically, I am comfortable working in Python and have experience using Git, machine learning frameworks, SQL, and data engineering workflows. My master's thesis involves developing multimodal machine learning systems, requiring substantial software development, experimentation, and management of complex datasets. I am particularly interested in the combination of machine learning models, healthcare registries, and simulation systems described in the CARE-AI project because it reflects the type of applied research environment in which I would like to develop further.